

Congruent Triangles

US Common Core – High School Geometry

Domain: Congruence (G-CO) | Cluster: Understand congruence in terms of rigid motions;
Prove geometric theorems

1 Triangle Angle-Sum Theorem

Theorem

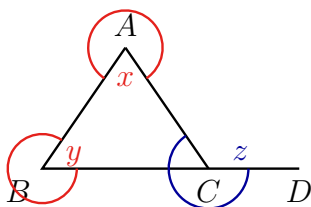
The sum of the interior angles of a triangle is equal to 180° .

2 Exterior Angle Theorem

Theorem

The measure of an exterior angle of a triangle equals the sum of the two remote (non-adjacent) interior angles.

$$x + y = z$$



Worked Examples

Example 1

Interior angles 40° and $2x^\circ$; exterior angle $(x + 53)^\circ$.

$$2x + 40 = x + 53 \implies x = 13^\circ$$

Example 2

Interior angles 40° and y° ; exterior angle 73° .

$$40 + y = 73 \implies y = 33^\circ$$

Example 3

Interior angles 40° and y° ; exterior angle 140° .

$$40 + y = 140 \implies y = 100^\circ$$

Example 4

Interior angles x° and 40° ; exterior angle $(x^2 + x + 20)^\circ$.

$$x + 40 = x^2 + x + 20 \implies x^2 = 20 \implies x = \sqrt{20} = 2\sqrt{5}^\circ$$

Example 5

Interior angles x° and 4° ; exterior angle $(x^2 + 2x + 6)^\circ$.

$$x + 4 = x^2 + 2x + 6 \implies x^2 + x + 2 = 0$$

Note: Example 5 as given in the source notes has a negative discriminant ($1 - 8 = -7$), so it has no real solution. This looks like a typo in the original figures (e.g. the exterior-angle expression or the 4° angle was likely meant to be a different value). Flagging rather than silently changing it – worth double-checking against the original lesson before this goes out to students.

3 Complementary and Supplementary Angles

Definitions

Two angles x° and y° are **complementary** if

$$x + y = 90^\circ,$$

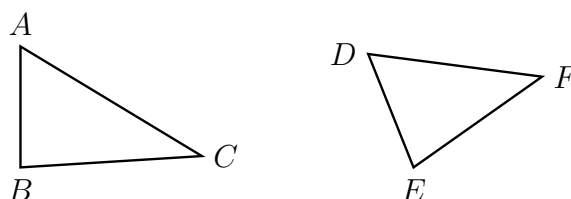
and **supplementary** if

$$x + y = 180^\circ.$$

4 Congruent Polygons

Definition

A triangle is a closed figure made of three line segments. Two triangles are **congruent** if one can be placed exactly on top of the other (superposition) – that is, all corresponding sides and angles are equal.



Theorem 1

If $\triangle ABC \cong \triangle DEF$, then their corresponding angles are congruent:

$$\angle A \cong \angle D, \quad \angle B \cong \angle E, \quad \angle C \cong \angle F.$$

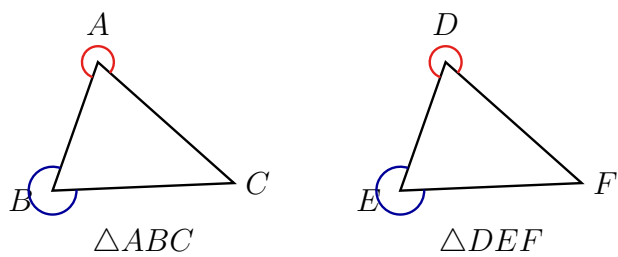
Theorem 2 (Properties of Congruence)

For triangles $\triangle ABC$, $\triangle DEF$, $\triangle XYZ$:

- **Reflexive:** $\triangle ABC \cong \triangle ABC$
- **Symmetric:** If $\triangle ABC \cong \triangle DEF$, then $\triangle DEF \cong \triangle ABC$
- **Transitive:** If $\triangle ABC \cong \triangle DEF$ and $\triangle DEF \cong \triangle XYZ$, then $\triangle ABC \cong \triangle XYZ$

Theorem 3 (Third Angle Theorem)

In $\triangle ABC$ and $\triangle DEF$, if $\angle A \cong \angle D$ and $\angle B \cong \angle E$, then $\angle C \cong \angle F$.

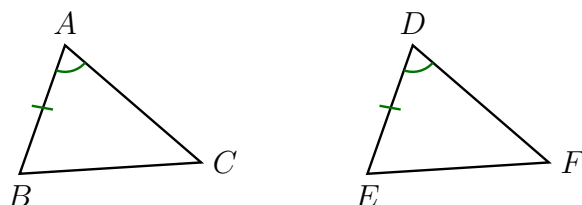


5 Triangle Congruence Postulates

(a) SAS – Side-Angle-Side

Postulate

If two sides and the included angle of one triangle are congruent to two sides and the included angle of another triangle, the triangles are congruent.



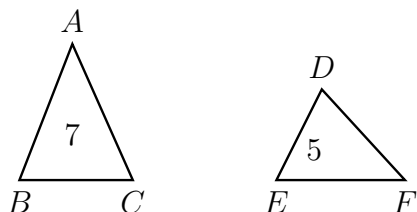
(b) SSS – Side-Side-Side

Postulate

If all three sides of one triangle are congruent to all three sides of another triangle, the triangles are congruent.

Why AA(A) alone is *not* enough

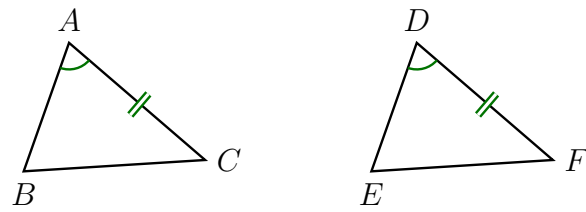
Two equilateral triangles with all angles 60° can still have different side lengths (e.g. side length 7 vs. side length 5) – so matching angles alone never guarantees congruence.



(c) AAS – Angle-Angle-Side

Postulate

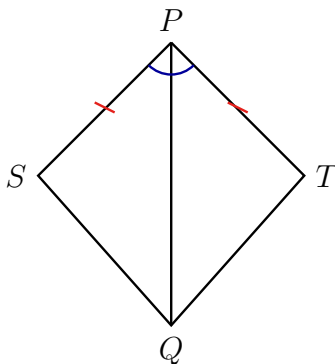
If two angles and a non-included side of one triangle are congruent to the corresponding two angles and side of another triangle, the triangles are congruent.



6 Worked Proof: Congruence by SAS

Setup

PQ bisects $\angle SPT$, and $SP \cong TP$. Prove $\triangle SPQ \cong \triangle TPQ$.



	Statement	Reason
S	$\overline{SP} \cong \overline{TP}$	Given
A	$\angle SPQ \cong \angle TPQ$	PQ bisects $\angle SPT$
S	$\overline{PQ} \cong \overline{PQ}$	Reflexive property (common side)
$\therefore$	$\triangle SPQ \cong \triangle TPQ$ (SAS)	